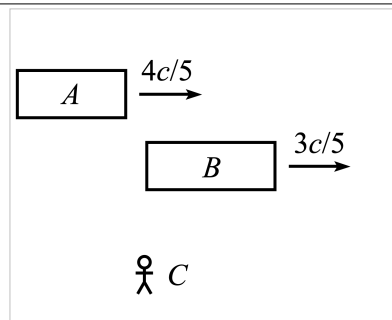


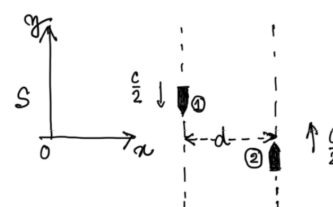
- Two trains, **A** and **B**, each have proper length L and move in the same direction. **A**'s speed is $4c/5$, and **B**'s speed is $3c/5$ with respect to ground. **A** starts behind **B** (see the figure).



- How long, as viewed by a person **C** on the ground, does it take for **A** to overtake **B**?
- At what speed, **B** would see **A** to cross?
- Explain the phenomena of **A** crossing **B** from the reference frame of **A**.

[By the *time of overtaking*, we mean time interval between the front of **A** passing the back of **B**, and the back of **A** passing the front of **B**.]

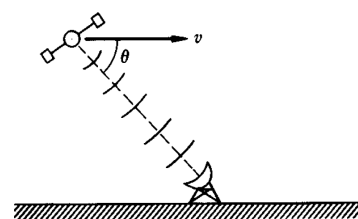
- As observed in an inertial frame S , two spaceships are travelling in opposite directions along straight, parallel trajectories separated by a distance d as shown in the figure. The speed of each ship is $c/2$.



- At the instant when the ships are at the points of closest approach, ship-1 ejects a small package which has speed $3c/4$ as viewed from S . From the point of view of the observer in ship-1, at what angle must the package be aimed for it to be received by ship-2?

- What is the speed of the package as seen by the observer in ship-1?

- Consider a Satellite which broadcasts a radio beacon signal (continuous signal), is moving with a velocity v at an angle θ with respect to a ground tracking station (see the figure beside), where the angle θ is measured in the rest frame of the observer. Find the frequency ν_D observed by the station if the frequency of the signal emitted is ν_0 from the rest frame of the Satellite.



- The kinetic energy (K) and the momentum (p) of a particle deduced from measurements on its track in nuclear experiment are 250 MeV and 368 MeV/c, respectively. Determine the rest mass of the particle.

-
5. A π meson decays into a muon (μ) and a neutrino (ν_μ) via the following reaction. The neutrino has zero rest mass.

$$\pi^+ \rightarrow \mu^+ \nu_\mu.$$

- (a) Using the relationship between total relativistic energy and momentum, show that the magnitude of the momentum p of the decay products in the pion rest frame is given by $p/c = (M_0^2 - m_0^2)/2M_0$; where M_0 is the pion rest mass and m_0 is the muon rest mass.
- (b) Consider the decay life time to be 2.5×10^{-8} sec in the rest frame of pion. Assuming that the π meson has kinetic energy equal to its rest energy, what distance would the meson travel before decaying as seen by an observer at rest?